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نظري

كلية الهندسة المعلوماتية

السنة الثانية

Signals



د. عماد الدين محمد

محتوى مجاني غير مخصص للبيع التجاري

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الاتصالات الرقمية

## Course Content:

- Signal
- Signal Classification
- Fundamental Signals
- Operations on Signals

In the last lecture we talked about signals and we will continue talking about them.

## Signals:

Signals are something that carries information.

This something is a physical quantity. **which means it can be:**

Stored

Manipulated

Transmitted

And/Or

- EX:** Sound is a Signal

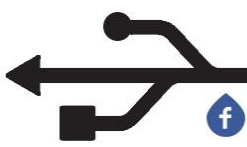
## Communications over net among applications:

We repeat **Signals** are something that carries information from applications.

These applications that send data are called a: **Source**, they send data to another application which we call **Destination**.

- Examples** of this so I can represent the signal: **I, V, U**.

we have **S**: source and **D**: destination.



As engineering students 🧐, we always use math to represent signals .

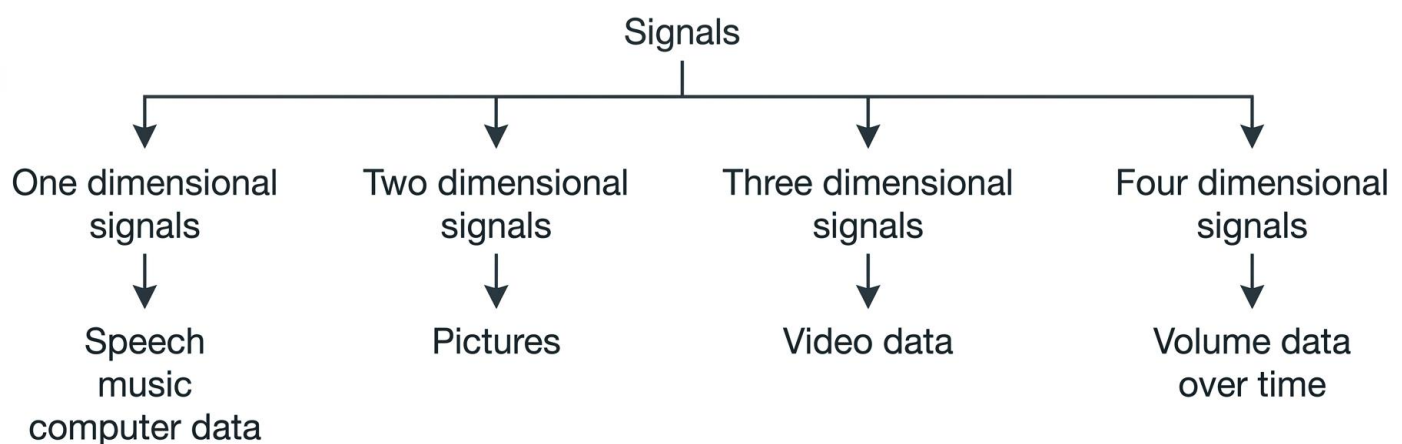
Mathematical representation is related to signal type:

1.  $X(t)$  continuous time signal.
2.  $X[n]$  discrete time signal.

**Continuous signals** are defined for a continuum of time, while **discrete signals** are defined only at specific discrete points in time .

The sign  $X(t)$  is continuous because there is  $t$ . However,  $X[n]$  is discrete because we have samples which takes specific values at certain moments. this means I don't know anything about the values unless they are told.

## Signals and Dimension



**Note:** three dimensions is: length-width-movement

There are applications that use systems working with a specific type of signals .. so we will classify the signals .

## Signals Classification

For example, when I'm working with radio communication, I'm dealing with signals that are different from the signals I need in biomedical imaging.

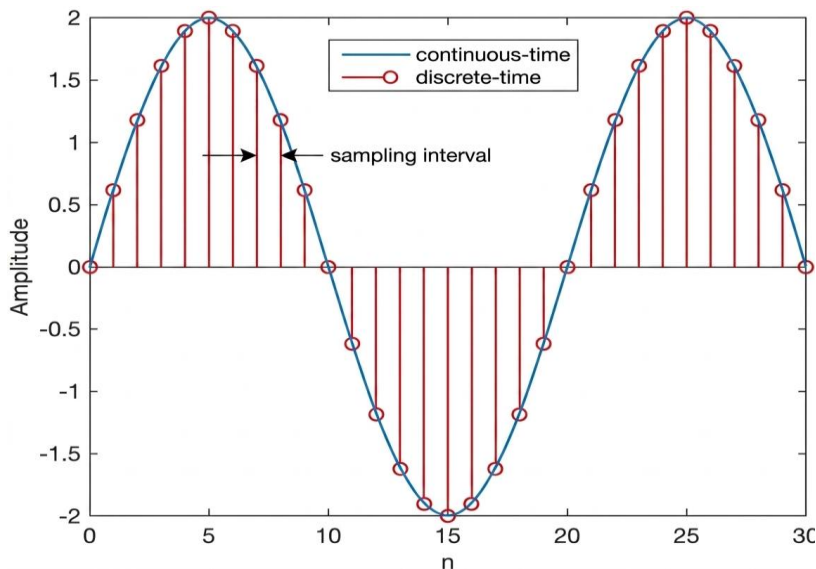
So we have standards by which these signals are classified

1) According to Time

1- Continuous-time VS Discrete-time:

Real-world continuous examples include temperature, sound waves, and wind speed;

Real-world discrete examples include daily stock prices, monthly sales, or digital audio;



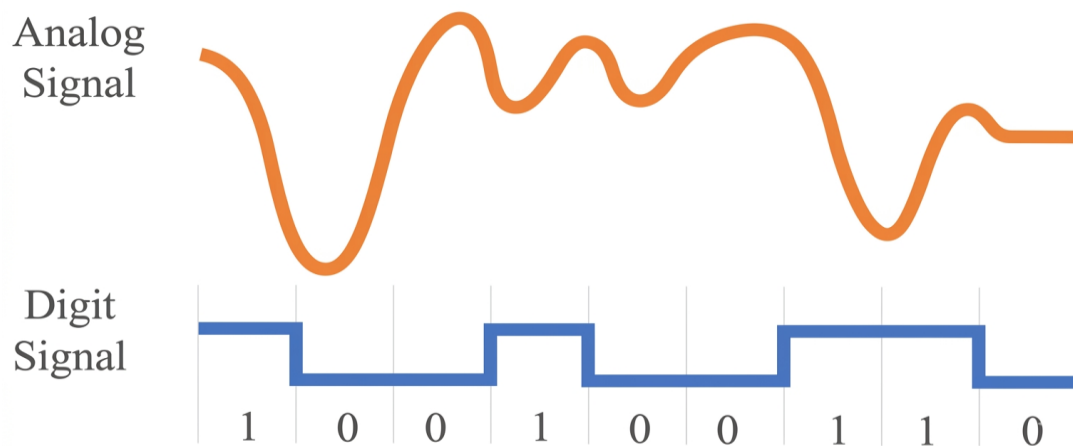
2- Analog VS Digital :

Digital signals is a special case of discrete-time signals by taking either 1 or 0;

Analog signals is a special case of continuous-time signals in every moment we have a specific value  $x(t)$  only on that range.

| Analog                                                                          | Digital                                                                                                                |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Analog signals are continuous in time and amplitude                             | Digital signals are discrete in both time and amplitude                                                                |
| Wave-like, and represent data through varying amplitude                         | Are discrete, binary, and step-based (0's and 1's)                                                                     |
| Analog is ideal for natural, real-world measurements Like sound and temperature | Digital excels in precision, storage, and noise resistance making it standard for modern computing, Data storage ..etc |

“Just finish it so you can complain about it properly 🤖”



### 3- Periodic VS Aperiodic:

**Periodic signals** repeat their pattern at regular, fixed time intervals.

**Aperiodic signals** (non-periodic) signals do not.

| Periodic                                                                | Aperiodic                                    |
|-------------------------------------------------------------------------|----------------------------------------------|
| I know the value it will be, why??<br>Because it will repeat it's own . | It doesn't repeat itself in a fixed pattern. |
|                                                                         |                                              |

Where do we use?

#### 1) Analog Periodic signals:

- Radio broadcasting (AM/FM).
- Sensor Systems...etc.

#### Information:

**FM:** Frequency Modulation

**AM:** Amplitude Modulation

**Modulation:** It is the process in which the original signal is taken and placed on a carrier wave in order to transmit it.

## 2) Analog Aperiodic signals:

-In communication systems for high rate transmission, natural and geophysical monitoring.

## 3) Discrete Periodic signals:

- DSP (Digital Signal Processing).
- Communication Systems for sampling.
- Signal generation (sin wave).
- Data Conversion.

## 4) Discrete Aperiodic Signals:

-Also DSP(digital filters).



### Example :

To eliminate unwanted frequency components (such as noise) from a discrete signal, a digital filter is applied to attenuate those specific frequencies while allowing others to pass

- Control Systems (stability).
- Speech and image processing.

## 2) According to Values (Predictability):

### 1- Deterministic VS Random:

| <u>Deterministic</u>                                                                                          | <u>Random</u>                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deterministic signals can be modeled exactly by a mathematical formula.<br><b>Ex:</b> $X(t)=5 \sin(\omega t)$ | Random (stochastic) signals have unpredictable behavior and are modeled statistically.<br><b>Ex:</b> Thermal noise, speech signals and seismic activity(Earthquakes). |

### 2- Even VS Odd:

| <u>Even</u>        | <u>Odd</u>        |
|--------------------|-------------------|
| $X(t)=X(-t)$       | $X(t)=-X(-t)$     |
| <p>Even signal</p> | <p>Odd signal</p> |

signal processing to simplify analyses like **Fourier** Transforms and decompose complex signals.

3) According to Behavior:

- **Energy & Power signals: (important)**

| <u>Energy signals</u>                                                              | <u>Power signals</u>                                                                |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <div>Have:</div> <div>Finite total energy and zero average power<br/>(Pulse)</div> | <div>Have:</div> <div>Finite power and infinite energy<br/>(Periodic signals)</div> |

$$p = \lim_{t \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$= \lim_{t \rightarrow \infty} \frac{1}{2T} \cdot \lim_{t \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt$$

Power Triangle

Note:

- We don't want to study the triangle now; we'll study it in future lectures.

- Mathematical equations are an exam question and are required to be memorized.

- To learn more, read the conclusions and understand the origin of the law.

| <u>Power signals</u>                                                                                             | <u>Energy signals</u>                                                |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1- Normalized average power is finite and non zero ( $0 < p < \infty$ ).                                         | Normalize average energy is finite and non zero ( $0 < E < \infty$ ) |
| 2- Power of CT signal: $p = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)^2 dt$ | Energy of CT signal: $E = \int_{-\infty}^{+\infty}  x(t) ^2 dt$      |
| 3- Power of DT signal : $p = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N  x(n) ^2$                  | Energy of DT signal : $E = \sum_{n=-\infty}^{\infty}  x(n) ^2$       |
| 4- Practical periodic signals are power signals.                                                                 | Non periodic signals are energy signals.                             |

### Power signals :

- **Periodic Signals:** Any repeating waveform (sine, cosine, square, triangular) that continues indefinitely.
- **Constant Signals:** A non-zero signal that lasts forever, such as a DC signal.
- **Step Function:** A signal that changes value at a specific time and stays there (  $u(t)$  ).

### Energy signals :

- **Pulses:** A single pulse or impulse that occurs once and dies out.
- **Non-periodic Signals:** Signals that do not repeat and exist only over a limited time intervals, such as a single decaying exponential (  $e^{-at}u(t)$  ).
- **Data and Images:** Images, audio and similar data packets usually contain a finite amount of energy.

## Fundamental Signals

Just like I have an alphabet of letters in Arabic that I use to form words, and from those words I build texts. Similarly, when we talked about **Signals**, there is a specific type of these signals that acts like an alphabet for **digital communications**.

**Fundamental signals** are basic, elemental waveforms—such as sinusoids, steps, impulses, and ramps — used to model and analyze complex physical systems and data.

These signals, often categorized by **time continuity** (continuous vs. discrete) or **predictability** (deterministic vs. random), are **essential** for understanding, representing, and manipulating data across time and frequency domains.

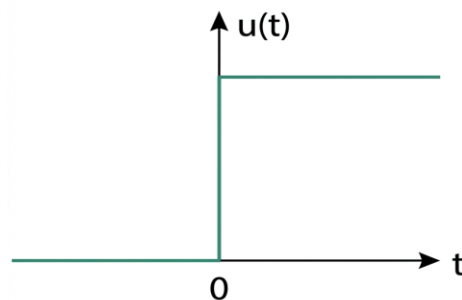
## Fundamental continuous-time signals (functions)

- 1- Unit step functions  $u(t)$  Heaviside function.
- 2- Unit Impulse function (Dirac Delta).
- 3- Unit Ramp function.
- 4- Exponential Signal.
- 5- Sinusoidal Signals.
- 6- Rectangular Pulse function.
- 7- Sinc function.

⇒ They are defined for all (t) in a continuous interval and can take any value, not just discrete values.

**“We don’t see things as they are, we see them as we are”**

## 1) Unit Step Function $u(t)$ :



$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

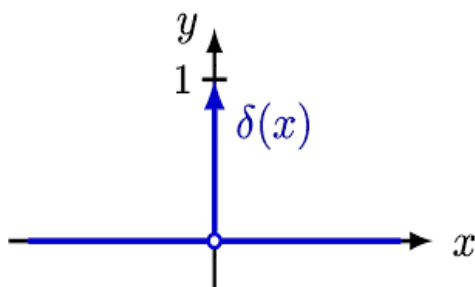
The **unit step function**, or Heaviside step function  $u(t)$  is a piecewise function that equals (0) for negative arguments and (1) for positive arguments, representing a switch that activates at  $t = 0$ . Defined as  $u(t) = 0$  for  $t < 0$  and  $u(t) = 1$  for  $t \geq 0$ , it is widely used in engineering to model systems and calculate Laplace transforms.

**Usage:** Switching on, rectangular pulse.

Representing an abrupt change,  
Such as turning on a switch.

$$\leftarrow u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

## 2) Unit Impulse Function <Dirac Delta>:

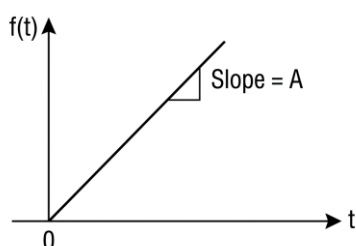


Dirac delta function

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0, & \text{if } x \neq 0 \end{cases}$$

**Usage:** physics, signal processing.

## 3) Ramp Function $R(t)$ :



$$\text{Basic Definition: } R(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

**Usage:** control systems, signal processing, power supply.

## Fundamental Discrete Signals:

Fundamental discrete signals are sequences defined at specific, distinct time intervals ( $n$ ) (where ( $n$ ) is an integer), essential for digital signal processing, control, and communication systems. Key elementary signals include the unit impulse, unit step, unit ramp, exponential, and sinusoidal signals. These are often derived by sampling continuous signals.

### 1) Unit Step Sequence:

#### Explanation:

-why sequence??

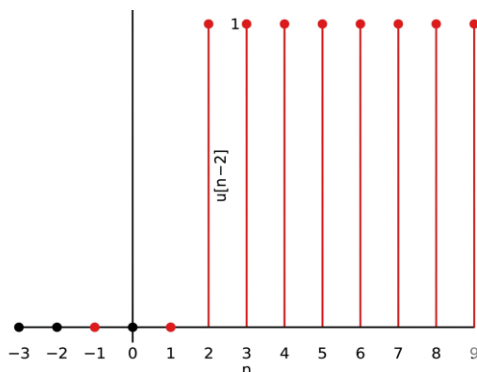
Because here I have moments (that have no value).

That means I cannot call specific moments  $u[n]$  instead of  $u(t)$ .

-How do I get values of discrete signals??

By doing samples from continuous time signal.

At a specific time, I call it **Sampling Period**, (as time passed I get the value and I do samples), Here we transform continuous time signal to discrete time signal.



$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

**Unit Step Sequence ( $u[n]$ ):** Defined as  $u[n] = 1$  for  $n \geq 0$  and  $u[n] = 0$  for  $n < 0$ . It represents the sudden application of a signal.

A unit step sequence, denoted as  $u[n]$ , is a fundamental discrete-time signal that acts as a step function, having a value of 0 for all negative indices ( $n < 0$ ) and a value of 1 for all non-negative indices ( $n \geq 0$ ). It is defined as  $u[n] = \{ \dots, 0, 0, \uparrow 1, 1, 1, 1, \dots \}$  where the arrow indicates the sample at  $n = 0$ .

-How do we get discrete time signal values??

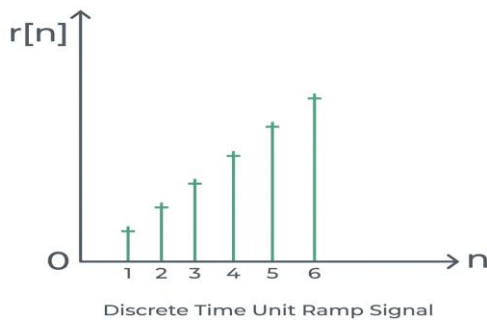


1] the professor gives me a math equation, I solve it and calculate the values at each point in time.

2] or it gives me all the information in the question, And you need to know where zero is, the arrow points to the value at zero.

**Applications:** Primarily used in signal processing and system analysis as a standard input test signal, particularly to determine a system's stability and response to sudden changes.

## 2) Unit Ramp Sequence:



$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

**Unit Ramp Sequence ( $r[n]$ )** is a discrete-time signal that starts at zero and increase linearly with a slope of one for non-negative indices.

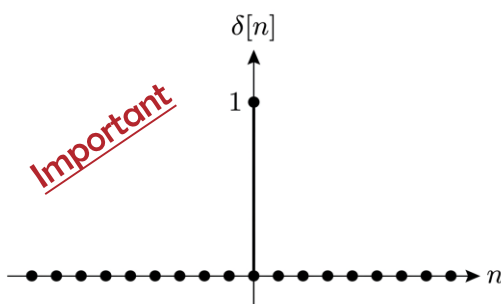
It is defined as  $r[n] = n \cdot u[n]$  which means  $r[n]=n$  for  $n \geq 0$ , and  $r[n]=0$  for  $n < 0$ . It acts as a foundational signal in digital signal processing and control systems.

### Common Usages:

-**Testing systems:** Used to check the response of digital systems to linearly increasing inputs.

-**Modeling:** Represents inputs that grow linearly over time.

## 3) Unit Impulse Sequence:



$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

**Unit Impulse Sequence:** A unit impulse sequence (or unit sample sequence), denoted as  $\delta[n]$ , is a fundamental discrete-time signal that equals 1 at  $n = 0$  and 0 for all other values of  $n$ .

It acts as an impulse at the origin and is essential for system analysis, convolution, and defining system impulse responses.

#### Usages;

**System Testing:** It is used as a standard input to observe transient responses and analyze the frequency response of filters.

**Signal Sampling:** The unit impulse function is used to model the ideal sampling process, creating a discrete sequence from a continuous signal.

**Signal Processing & Edge Detection:** In image processing, it can be used to model spatial characteristics.

### Inter-relationships:

**Note:** Relationships to memorize + read their conclusion

#### ↙ Continuous Time:

- $u(t) = \int_{-\infty}^t \delta(\tau) d\tau$
- $\delta(t) = \frac{du(t)}{dt}$

“In the middle of chaos,  
there is quiet meaning”

#### ↙ Discrete Time:

- $u[n] = \sum_{k=0}^{\infty} \delta[n - k]$
- $\delta[n] = u[n] - u[n - 1]$

#### ↙ U(t) & R(t):

- $r(t) = \int_{-\infty}^t u(t) dt = \int_{-\infty}^0 u(t) dt + \int_0^t u(t) dt = 0 + \int_0^t u(t) dt = t;$

$$r(t) = \begin{cases} 0 & \text{for } t < 0 \\ t & \text{for } t > 0 \end{cases}$$

#### ↙ U[n] & R[n]:

- Unit Step Sequence ( $u[n]$ ):

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- Unit Ramp Sequence ( $r[n]$ ):

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$r[n] = n \cdot u[n]$$



$$r[n] = \sum_{k=0}^n u[k]$$

$$u[n] = r[n] - r[n-1]$$

- Unit Impulse Sequence:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

- Unit Step Sequence:

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- Unit Ramp Sequence:

$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

### ⚡ From Impulse to Ramp (Accumulation):

- First summation (Impulse to Step): The running sum of an impulse is a step function.

$$\sum_{k=-\infty}^n \delta[k] = u[n]$$

- Second summation (Step to Ramp): The running sum of a step function is a ramp function.

$$r[n] = \sum_{j=-\infty}^n u[j] = \sum_{j=0}^n \left( \sum_{k=0}^j \delta[k] \right)$$

- Therefore:

$$r[n] = \sum_{k=0}^n u[k]$$

“The unseen shapes more than the visible”

### From Ramp to Impulse (Difference):

- First Difference:  $r[n] - r[n-1] = u[n]$  (the slope changes only at  $n=0$ ).
- Second Difference:  $u[n] - u[n-1] = \delta[n]$ .
- Substituting  $u[n]$  gives the relationship:

$$\delta[n] = r[n] - 2r[n-1] + r[n-2]$$

| Signal Relationship | Formula                                      |
|---------------------|----------------------------------------------|
| Impulse to step     | $u[n] = \sum_{k=0}^n \delta[k]$              |
| Step to Ramp        | $r[n] = \sum_{k=0}^n u[k]$                   |
| Impulse to Ramp     | $r[n] = \sum_{k=0}^n \sum_{m=0}^k \delta[m]$ |
| Ramp to Impulse     | $\delta[n] = r[n] - 2r[n-1] + r[n-2]$        |

### Operations on signals:

Operations cover : 1) time. 2) amplitude.

#### Time:

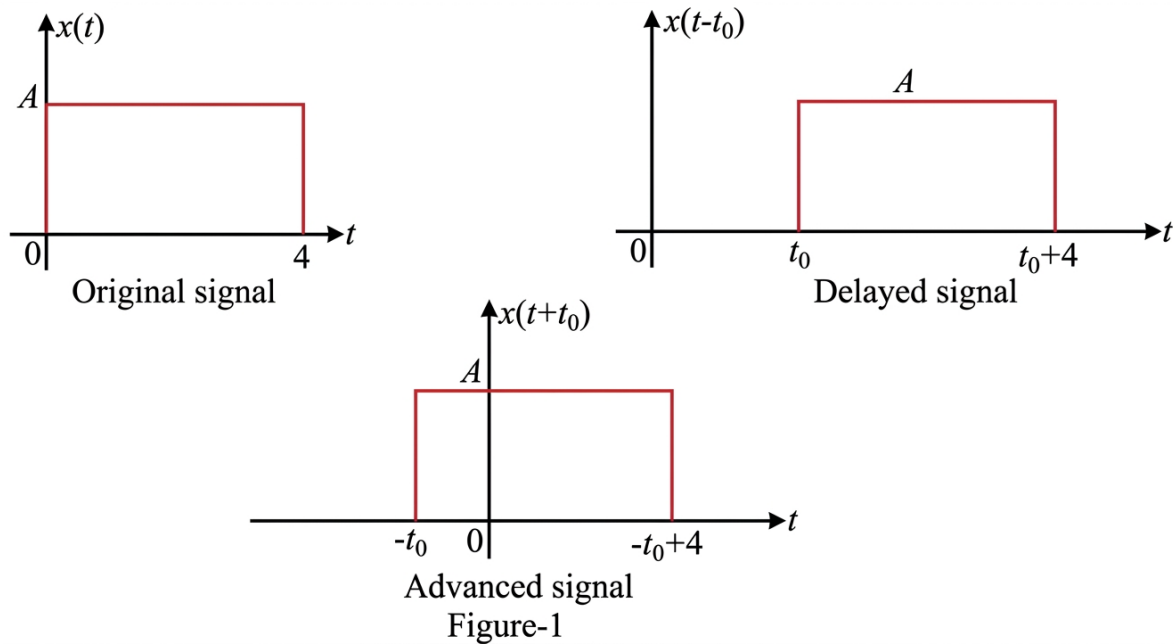
- Shifting
- Scaling
- Reversal

**Time Shifting:** Moves a signal  $x(t)$  forward or backward on the time axis by a constant  $t_0$ , represented as  $y(t) = x(t \pm t_0)$ , without changing its shape or amplitude.

**Negative Shift ( $t - t_0$ ):** Delays the signal, moving it right (time delay).

**While a Positive Shift ( $t + t_0$ ):** Advances it left.

“Sometimes the question is more important than the answer”

**Note:**

How do I use the signal before it comes??

Physically, incorrect.

**The solution:**

We have values of the signals, we take one of these values and treat it as if it were the actual value, and the system proceeds until the true value arrives.

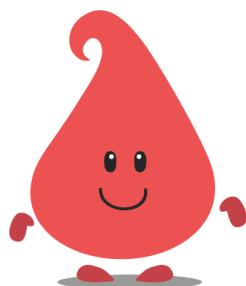
Then, the system performs a comparison between the received value and adjusts itself accordingly to match that value.

-We talk about Random Signal and Deterministic values.

**Time shifting** is a fundamental operation in signal processing that involves moving a signal **forward** (delay) or **backward** (advance) along the time axis. This operation is a core building block for various engineering and analytical applications.

**Why??**

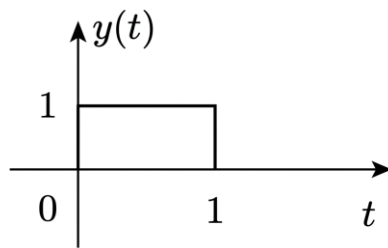
- **Synchronization and Alignment:** It is used to align multiple audio or data tracks so they occur simultaneously, which is critical in multimedia production and multi-sensor data fusion.



## Time scaling:

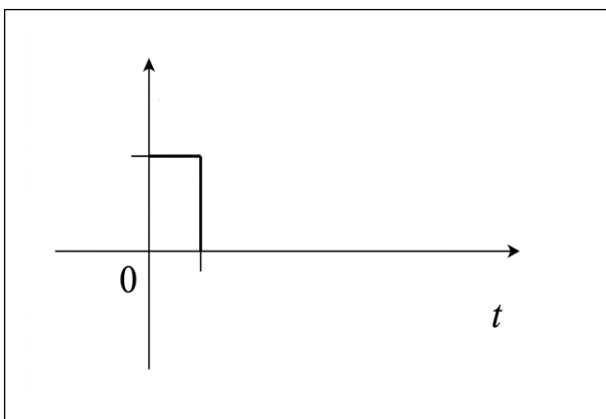
Time scaling is a signal processing operation that compresses or expands a signal  $x(t)$  by multiplying its time variable by a non-zero constant  $\alpha$ , resulting in  $y(t) = x(\alpha t)$ . If  $|\alpha| > 1$ , the signal is compressed (**speeds up**), and if  $0 < |\alpha| < 1$ , it is expanded (**slows down**), while the amplitude remains unchanged.

**Ex:** given  $y(t)$ , find  $w(t)=y(3t)$  and  $v(t)=y(t/3)$



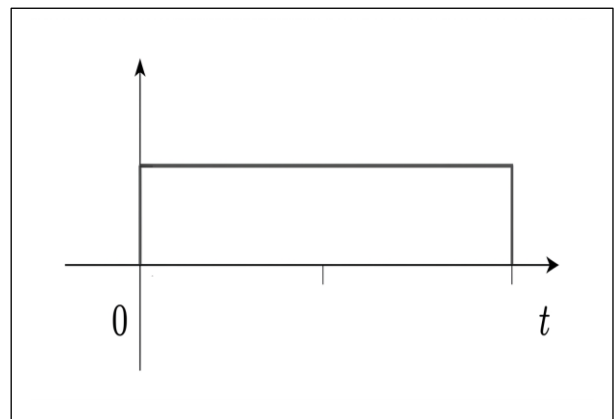
The solution:

$$w(t) = y(3t)$$



Compression

$$v(t) = y(t/3)$$



Expansion

**Homework:** why? → Adjusting the time scale of a signal according to the application.

## Time shifting & scaling:

Combining time scaling and time shifting, represented by a transformation like:

$$y(t) = X(at - b)$$

1. **Scale First:** Perform the time scaling ( $t \rightarrow at$ ) first.
2. **Shift Second:** Perform the time shift ( $at \rightarrow at - b$ ) second.

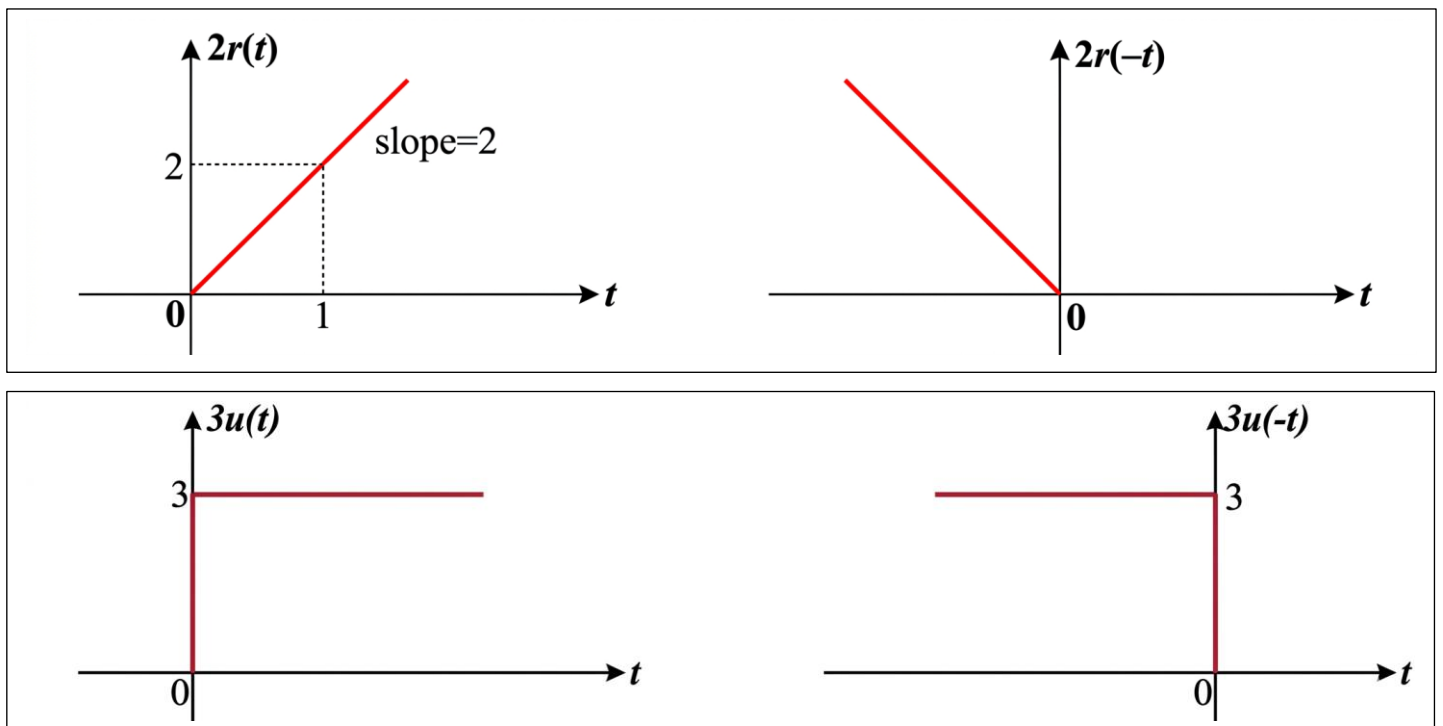
For example, to produce  $x(3t - 6)$ :

- Start with  $x(t)$ .
- Scale it to  $x(3t)$  (compress by 3).
- Then shift it by 6 to get  $x(3(t - 2))$ , which is  $x(3t - 6)$ .

## Time Reversal:

Time reversal of a ramp signal,  $r(t) = t \cdot u(t)$ , involves flipping the signal across the vertical axis (amplitude axis) to create a mirror image, resulting in  $r(-t) = -t \cdot u(-t)$ .

this means the signal, which originally starts at  $t = 0$  and increases linearly for positive time, now exists only for negative time and decreases towards zero as time moves towards zero.



**“THE END”**